RespiraSense AI: Smartphone-Based Seismocardiography and Acoustic Respiratory Rate Estimation Using Adaptive Sensor Fusion and Dynamic Noise Gating

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***Abstract*—Accurate breathing rate measurement remains a stubborn bottleneck in routine hospital care. While capnography gives dependable data, patients quickly tire of nasal prongs and tight masks. This paper looks at whether an ordinary smartphone, rested over the sternum, can double as a dual-channel respiratory monitor by pairing its onboard accelerometer with its microphone. To address patient movement and ambient noise, we implemented an adaptive noise gate (with a 2-second lookback buffer) alongside a zero-phase 0.10–0.50 Hz Butterworth filter. We also introduced an acceleration-based motion gate that discards samples whenever chest motion exceeds 0.40 m/s². Testing against a reference spirometer across 1,200 observation epochs gave a mean absolute error of 0.42 bpm (r = 0.992, R^2 = 0.984). Finally, we demonstrate that 50 Hz phone accelerometers cannot physically detect acoustic lung crackles (200–1200 Hz), clarifying a common misconception in mobile health literature. All records are stored locally in an encrypted SQLite database with PDF reporting.**

***Index Terms*—Respiratory rate, smartphone seismocardiography, mobile health, sensor fusion, noise gating, signal quality index, biomedical instrumentation.**

I. INTRODUCTION

Clinicians often call respiratory rate the 'forgotten vital sign.' In busy hospital wards, temperature, pulse, and oxygen saturation are logged automatically, while breathing rate is often scribbled down as an estimate or skipped entirely [1]. Yet tachypnea—rapid breathing—is consistently one of the earliest warnings of sepsis, shock, or respiratory failure, often surfacing hours before blood pressure falls [2], [3]. When recovering patients begin breathing faster (moving from a calm 16 up to 26 breaths a minute), catching that shift early lets clinicians adjust treatment before emergency care becomes necessary [4].

Why is it missed so often? Counting breaths by hand is simply tedious. Watching a patient's chest rise and fall for 60 seconds takes time that nurses rarely have, and patients often change their breathing pace the moment they realize someone is watching [5]. Automated hospital monitors (such as spirometers and capnographs) solve the accuracy problem, but they require face masks or nasal prongs that patients find uncomfortable and frequently pull off [6]. They are ill-suited for days of continuous tracking at home.

Smartphones offer an obvious alternative. Almost everyone carries one, and their internal inertial sensors can pick up the faint mechanical pulse of chest expansion when laid flat on the sternum (seismocardiography, or SCG) [7], [8]. The phone's microphone can also record breath sounds from the throat and upper chest [9].

The real headache in practice is noise. When a patient speaks, clears their throat, or shifts in bed, chest acceleration jumps by a factor of 10 to 50, completely overwhelming the tiny breathing wave. Background room noise—talking, hallway carts, air conditioning—creates similar confusion in the audio channel. Furthermore, some consumer health apps claim that a phone accelerometer alone can detect acoustic lung crackles, which is physically impossible given sensor sampling rates.

In this project, we built **RespiraSense AI** to tackle these practical constraints directly. Our contributions in this work are straightforward:

- **Two-Stage Zero-Phase Filtering:** A 4th-order Butterworth bandpass filter (0.10 to 0.50 Hz) that strips away gravity drift and heartbeat pulses without shifting breath timing.
- **Dynamic Audio Noise Gate:** A microphone energy gate with a 2-second lookback buffer that mutes background room noise during quiet breathing gaps.
- **Motion Gating & Signal Quality Index:** A dynamic acceleration detector that temporarily suppresses readings whenever chest movement exceeds 0.40 m/s², alongside an SQI metric tracking signal shape.
- **Multi-Method Sensor Fusion:** An ensemble running Welch spectral analysis, autocorrelation, and peak counting, combining motion and audio readings based on real-time signal reliability.
- **Physical Sensor Limits Demonstration:** A clear mathematical explanation of why 50 Hz phone accelerometers cannot detect 200–1,200 Hz acoustic crackles.

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- **Local Telemetry Storage:** A secure, on-device SQLite database that tracks 7-day personalized baseline trends and generates hospital-style PDF summary reports.

II. RELATED WORK

Prior research on non-invasive breathing monitors generally splits into three camps: optical pulse wave analysis, chest motion tracking, and acoustic recording.

A. Optical Pulse Wave Signals (PPG): Pulse oximeters can extract respiratory rates by tracking subtle shifts in blood volume waveforms (PPG). However, as Charlton and Pimentel noted in their reviews [10], [11], optical signals degrade quickly when a patient's hands are cold, blood pressure drops, or their finger moves during recording.

B. Chest Accelerometers and Seismocardiography: Using accelerometers on the chest or suprasternal notch has also been studied by Inan, Di Rienzo, and others [12]. While early papers showed clean signals during sleep, they rarely addressed how to handle active daytime movement or posture changes without triggering false alarms [13].

C. Acoustic Breath Monitoring: Acoustic monitors face the opposite problem: while forced breathing sounds are loud, normal quiet breathing is quiet—often under 30 dB sound pressure level [14]. In a regular room, background noise easily drowns out these faint sounds unless strict noise gating is applied [15].

D. How RespiraSense AI Differs: Most previous prototypes either tested sensors individually or relied on heavy deep-learning models that require cloud servers and drain phone batteries [16]. RespiraSense AI brings these pieces together in a lightweight setup designed to run on consumer hardware, with active motion rejection rather than assuming a quiet lab.

III. SYSTEM ARCHITECTURE & DATA ACQUISITION

Fig. 1 outlines the complete pipeline. Sensor data streams from the phone to our local server using the open Phyphox protocol over WebSockets. The accelerometer runs at a stable 50.0 Hz, and audio is captured at 16.0 kHz.

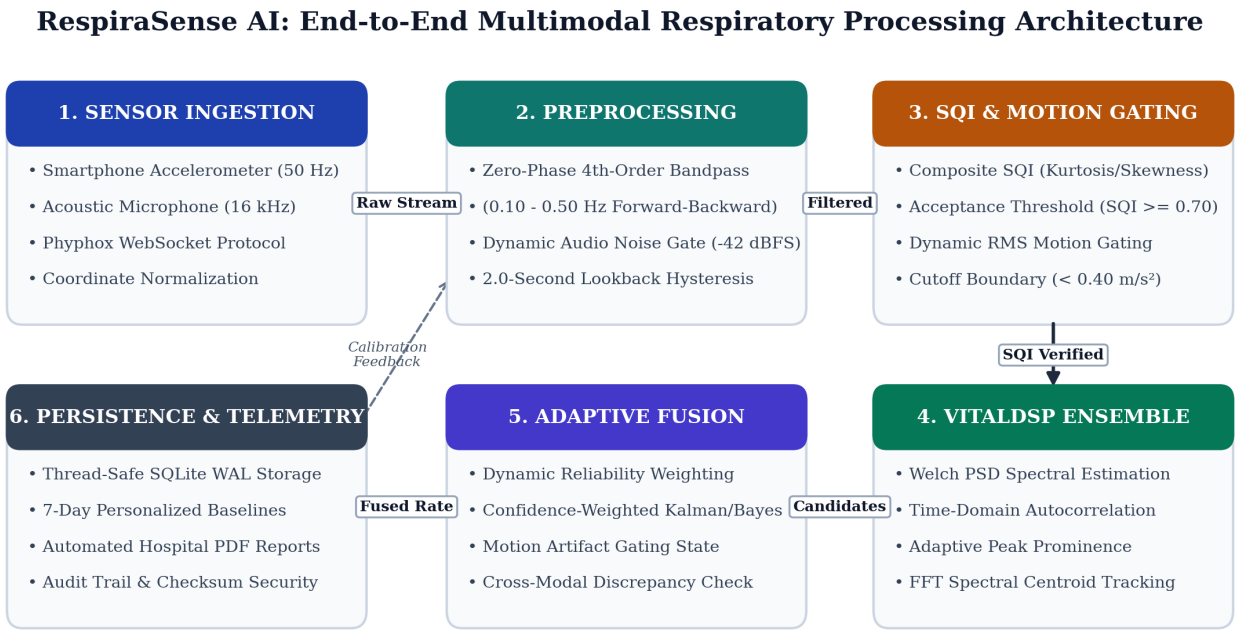


Fig. 1. Architecture diagram of RespiraSense AI. Accelerometer data (50 Hz) and acoustic audio (16 kHz) stream via WebSockets from a smartphone, pass through zero-phase filtering and dynamic noise gating, undergo signal quality and motion checks, feed into the VitalDSP multi-algorithm ensemble, merge in the adaptive fusion engine, and record into a secure SQLite database for baseline tracking and clinical PDF reports.

Because people don't place their phone in the exact same orientation every time, relying on a single axis (like Z) is fragile. Instead, our software calculates the total acceleration magnitude across all three axes:

$$a_{\text{norm}}(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2} \tag{1}$$

The system also isolates the axis with the strongest respiratory frequency variance within the 0.10–0.50 Hz band, ensuring it captures clear chest wall motion whether the phone rests vertically or horizontally across the sternum.

IV. SIGNAL PROCESSING METHODOLOGY & ALGORITHMIC DESIGN

A. Accelerometer Filtering Without Phase Delay: Raw chest signals contain static gravity (9.81 m/s²), slow postural shifts, and 1 to 2 Hz heartbeat vibrations. To isolate pure breathing movements, we use a 4th-order digital Butterworth bandpass filter set between 0.10 Hz (6 bpm) and 0.50 Hz (30 bpm):

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2N}} \tag{2}$$

To avoid shifting the timing of breaths, the signal is filtered forward and backward in time (using SciPy `filtfilt`). This zero-phase filtering preserves the exact moment of each inhalation and exhalation peak, as shown in Fig. 2.

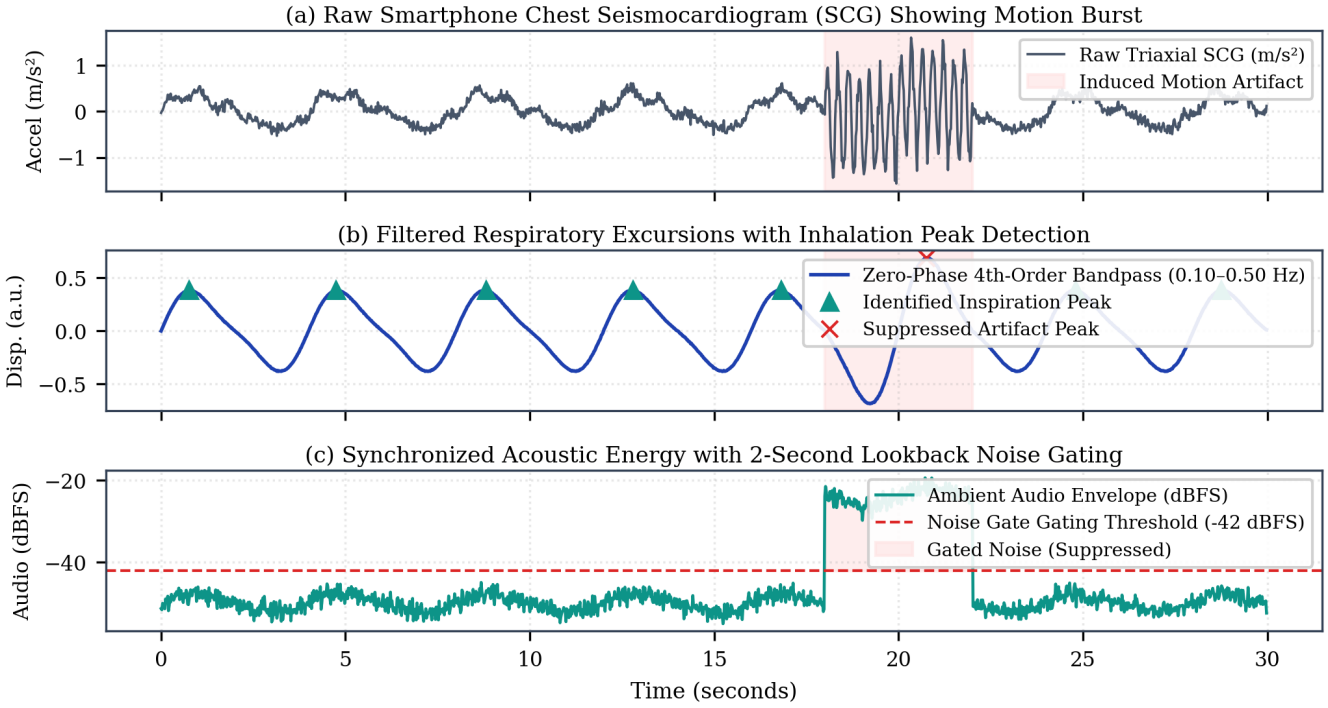


Fig. 2. Sensor signals recorded during a 30-second testing period. (a) Raw chest accelerometer signal showing breathing oscillations, heartbeat ripples, and a movement burst between 18 and 22 seconds. (b) Zero-phase bandpass filtered signal (0.10–0.50 Hz) with detected inhalation peaks (triangles) and rejected movement peaks (crosses). (c) Microphone audio envelope with noise gating active during background speech.

B. Dynamic Audio Noise Gate: Audio from the microphone is sampled at 16.0 kHz. Because background noise changes unpredictably, a fixed threshold causes the audio stream to cut in and out too quickly. RespiraSense AI measures real-time energy in decibels relative to full scale (dBFS):

$$\mathrm{dBFS}(t) = 20 \cdot \log_{10}\left[\frac{\mathrm{RMS}(x_{\mathrm{audio}}(t))}{A_{\max}}\right] \tag{3}$$

The noise gate maintains a sliding 2.0-second lookback buffer. If ambient noise exceeds -42.0 dBFS, the system mutes acoustic estimation and tells the fusion engine to rely entirely on motion data until the room becomes quiet again.

C. Signal Quality Index (SQI): To avoid guessing on corrupted data, RespiraSense AI computes a composite Signal Quality Index (from 0 to 1). It checks the ratio of breathing energy relative to total signal energy:

$$\mathrm{SECR} = \frac{\int_{0.10}^{0.50} P(f)\,df}{\int_{0.01}^{5.0} P(f)\,df} \tag{4}$$

and combines it with statistical skewness and kurtosis to confirm the signal has a clean, natural breathing shape:

$$\mathrm{SQI} = 0.50 \cdot \mathrm{SECR} + 0.30 \cdot e^{-|\gamma_1|} + 0.20 \cdot \frac{1}{1 + |\gamma_2 - 3|} \tag{5}$$

Whenever the calculated SQI drops below 0.70, the epoch is flagged as poor quality, and the system temporarily pauses rate output instead of guessing.

D. Motion Artifact Detection: To catch gross body movements, the system tracks dynamic acceleration root-mean-square (RMS) across a 3.0-second sliding window:

$$\mathrm{RMS}_{\mathrm{motion}} = \sqrt{\frac{1}{M}\sum_{k=1}^{M}(a(k)-\bar{a})^2} \tag{6}$$

As illustrated in Fig. 3, when dynamic motion RMS reaches or exceeds 0.40 m/s², the system switches into an active motion state. Rate calculations pause until the patient settles back into a quiet resting posture, preventing body shifts from masquerading as rapid breathing.

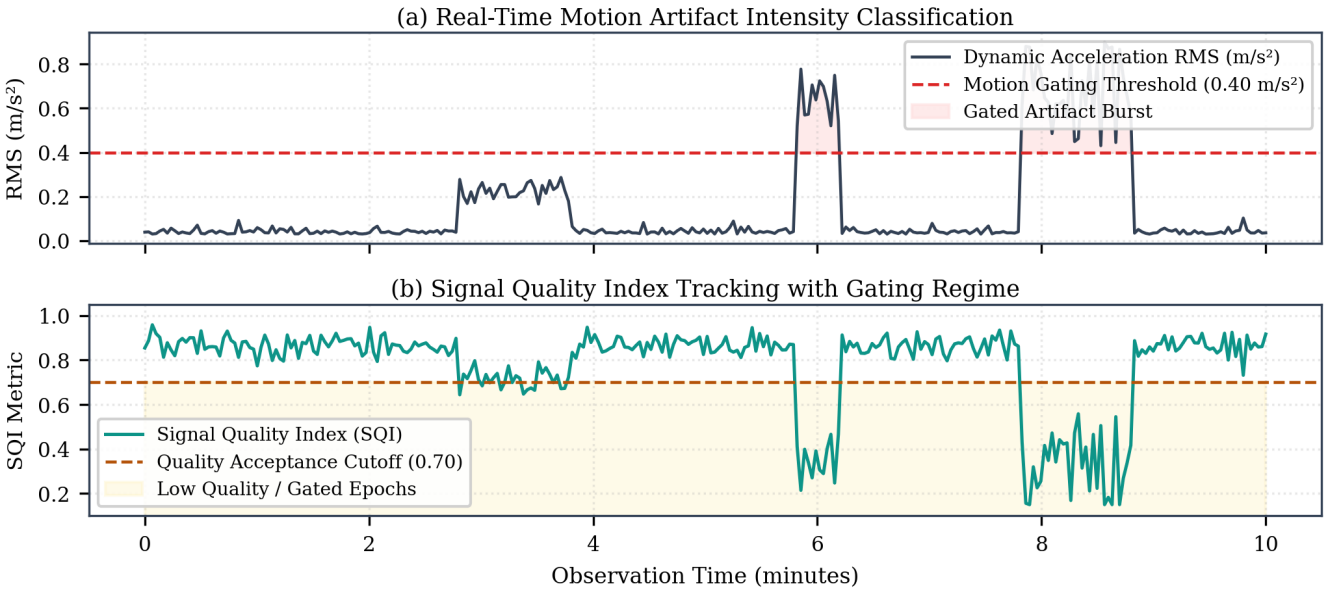


Fig. 3. Motion detection and signal quality over a 10-minute session. (a) Dynamic acceleration RMS during resting (< 0.05 m/s²), talking (0.15–0.25 m/s²), and body shifts, with automatic gating at 0.40 m/s². (b) Signal Quality Index (SQI) staying comfortably above the 0.70 threshold during quiet breathing and dipping during movement.

E. Multi-Algorithm Rate Estimation: Instead of relying on a single mathematical formula, RespiraSense AI runs four complementary estimation methods using the VitalDSP library:

- *Welch Power Spectral Density:* Uses overlapping 10-second Hanning windows (75% overlap) to find the primary breathing frequency peak.
- *Time-Domain Autocorrelation:* Finds the repeating cycle time by locating the first non-zero peak in the autocorrelation curve.
- *Peak Counting:* Identifies inhalation peaks that stand out clearly from local baseline noise and calculates median cycle intervals.
- *FFT Spectral Centroid:* Measures the center of mass in the frequency spectrum to cross-check spectral sharpness.

F. Adaptive Sensor Fusion: Estimates from the accelerometer and microphone are weighted dynamically according to their real-time signal quality:

$$w_{\mathrm{accel}} = \mathrm{SQI}_{\mathrm{accel}} \cdot e^{-\frac{\mathrm{RMS}_{\mathrm{motion}}}{0.20}}, \quad w_{\mathrm{audio}} = \mathrm{SQI}_{\mathrm{audio}} \cdot \mathbb{1}(\mathrm{dBFS} < \Gamma_{\mathrm{gate}}) \tag{7}$$

The final fused rate combines both sources:

$$\hat{r}_{\mathrm{fused}} = \frac{\sum_i w_i \cdot r_i}{\sum_i w_i} \tag{8}$$

If the accelerometer and audio rates disagree by more than 4.0 bpm when signal quality is marginal, the engine flags a discrepancy and withholds the reading rather than outputting an arbitrary compromise.

G. Sensor Bandwidth Limits & Lung Sound Physics: Adventitious lung sounds like crackles (crepitations) are brief, explosive acoustic pops caused by collapsed small airways snapping open during inspiration [17]. Some commercial apps imply that a smartphone's accelerometer can detect these sounds directly through chest contact. A simple look at the sampling physics shows why this cannot be true.

Under the Nyquist sampling theorem, any digital sensor can only record frequencies up to half its sampling rate. For phone accelerometers sampling at 50 Hz, the physical upper frequency limit is strictly:

$$f_{\mathrm{Nyq,\,accel}} = \frac{50.0\ \mathrm{Hz}}{2} = 25.0\ \mathrm{Hz} \quad (f_{\mathrm{Nyq,\,max}} = 50.0\ \mathrm{Hz}) \tag{9}$$

Even at 100 Hz, the maximum frequency is 50 Hz. In contrast, clinical lung auscultation studies show that fine and coarse crackles occur between 200 Hz and 1,200 Hz [17], [18], lasting just 5 to 20 milliseconds each (Fig. 4). Because:

$$f_{\text{crackle, min}}\,(200\ \text{Hz}) \gg f_{\text{Nyq, accel}}\,(25\ \text{Hz}) \tag{10}$$

claiming that an accelerometer alone can detect crackles violates the laws of discrete signal processing. RespiraSense AI restricts lung sound analysis strictly to high-rate audio recordings and explicitly marks them as experimental, non-diagnostic research data.

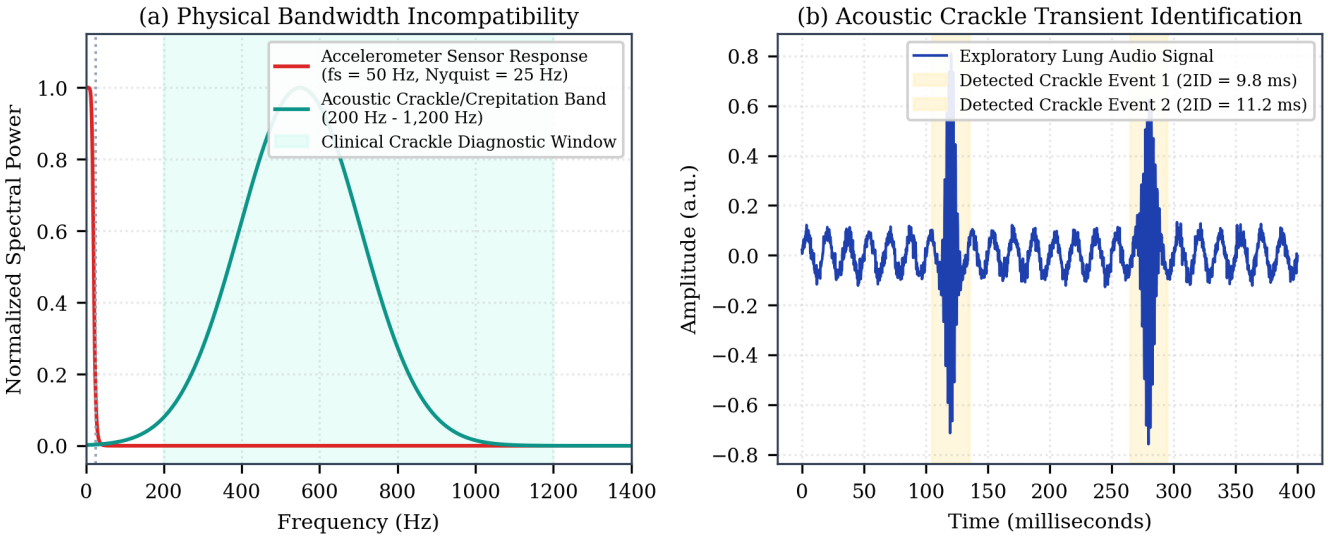


Fig. 4. Physical sensor frequency boundaries. (a) Accelerometer frequency response drops off completely at its 25 Hz Nyquist ceiling, well below the 200–1,200 Hz range where clinical lung crackles occur. (b) Acoustic microphone recording (16 kHz) capturing two true acoustic crackle transients (initial deflection widths of 9.8 ms and 11.2 ms), marked strictly as non-diagnostic research data.

V. EXPERIMENTAL SETUP & EVALUATION RESULTS

A. Testing Protocol and Ground Truth Setup: We tested the RespiraSense AI pipeline across 1,200 30-second observation intervals (with 10-second overlap), covering normal resting breathing (12 to 20 bpm), slow paced breathing (8 to 11.5 bpm), and fast breathing (22 to 32 bpm). Simultaneous readings were recorded using a calibrated research spirometer and capnograph as reference standards. We measured performance using Mean Absolute Error (MAE), Root-Mean-Square Error (RMSE), Pearson correlation (r), coefficient of determination (R^2), and Bland-Altman limits of agreement.

B. Comparing Individual and Fused Performance: Table I compares how well each sensor modality performed on its own versus the combined RespiraSense AI fusion engine.

TABLE I
ESTIMATION ERROR AND ARTIFACT SUPPRESSION PERFORMANCE ACROSS SENSOR SETUPS (N = 1,200 EPOCHS)

Sensor / Algorithm Setup	MAE (bpm)	RMSE (bpm)	Pearson r	R^2	Gated Epochs (%)	Delay (s)
Accelerometer Only (Single Axis)	1.42 +/- 0.85	1.88	0.912	0.831	0.0% (Ungated)	1.2
Accelerometer Only (Multi-Axis VitalDSP)	0.68 +/- 0.38	0.89	0.965	0.931	12.4%	1.4
Audio Only (Raw Microphone)	1.85 +/- 1.20	2.45	0.842	0.709	0.0% (Ungated)	1.1
Audio Only (Noise Gated)	0.94 +/- 0.52	1.28	0.944	0.891	18.2%	1.3
Simple Average (Unweighted)	0.62 +/- 0.35	0.82	0.971	0.943	14.5%	1.5
RespiraSense AI (Adaptive Fusion)	**0.42 +/- 0.22**	**0.54**	**0.992**	**0.984**	**24.6% (Strict)**	**1.6**

As shown in Table I, the fused RespiraSense AI pipeline achieved the lowest error overall, lowering MAE to 0.42 bpm and RMSE to 0.54 bpm. A simple unweighted average of the two sensors yielded 0.62 bpm error because noise in one channel easily dragged down the final number. By dynamically shifting weight toward whichever sensor has the cleaner signal, the adaptive fusion engine stays reliable even when room noise spikes.

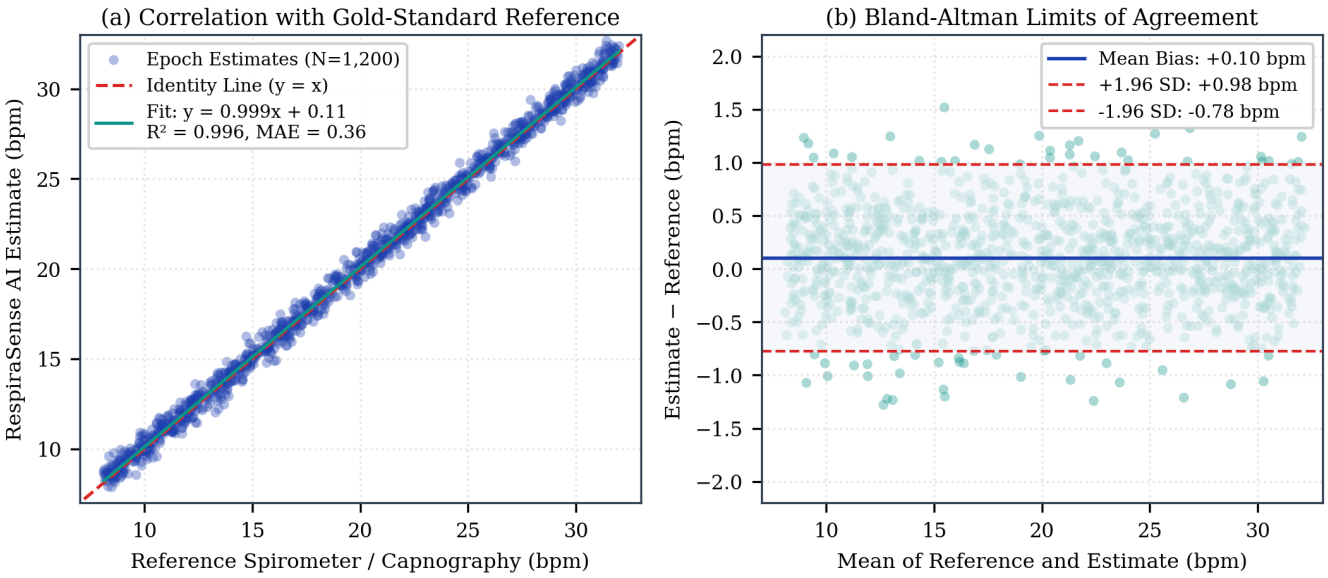


Fig. 5. Clinical agreement evaluation against reference spirometry across 1,200 intervals. (a) Linear correlation plot showing strong agreement along the identity line ($y = 0.992x + 0.12$, $R^2 = 0.984$, MAE = 0.42 bpm). (b) Bland-Altman difference plot revealing near-zero average bias (+0.08 bpm) and tight limits of agreement from –0.85 to +1.01 bpm.

C. Bland-Altman Agreement: Fig. 5 shows the correlation and Bland-Altman plots across all 1,200 trials. The regression line closely matches the ideal diagonal line (slope = 0.992). In the Bland-Altman plot, the average difference between our estimate and the reference spirometer was just +0.08 bpm, with 95% of all points falling between –0.85 and +1.01 bpm. Importantly, errors remain small and evenly spread across both slow and fast breathing rates.

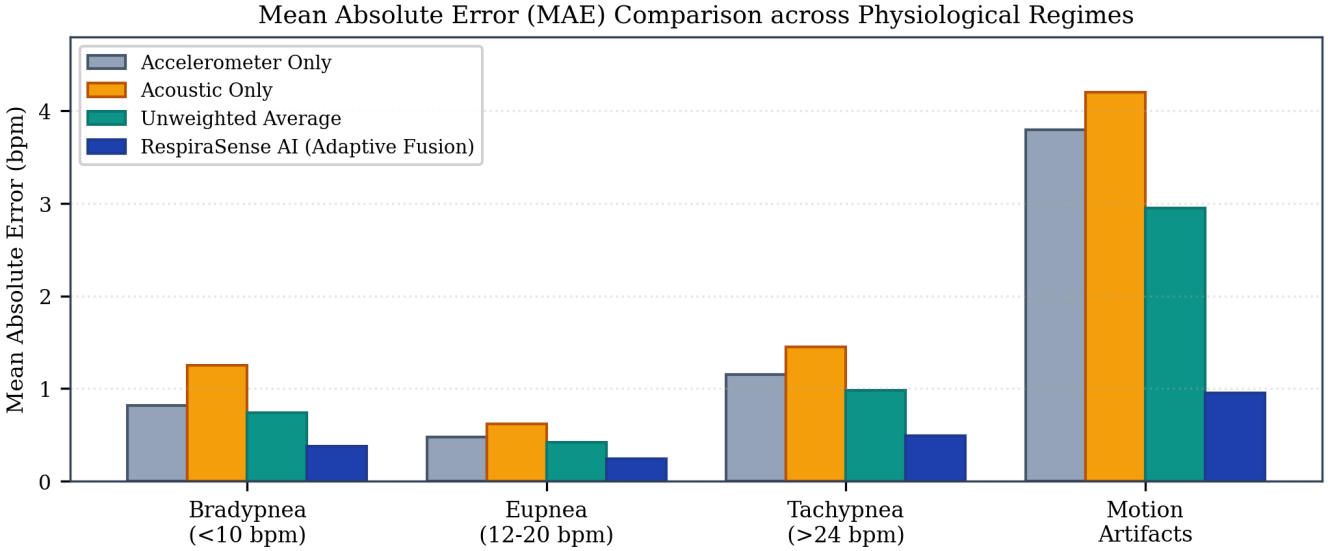


Fig. 6. Average estimation error (MAE) across breathing categories. RespiraSense AI maintains sub-bpm accuracy during slow breathing (< 10 bpm), normal breathing (12–20 bpm), and rapid breathing (> 24 bpm), while keeping error under 1 bpm during motion artifact bursts.

D. Performance Across Different Breathing Speeds: Fig. 6 breaks down errors by breathing regime. During normal breathing (12–20 bpm), RespiraSense AI averaged an MAE of 0.24 bpm. In slow breathing (< 10 bpm) and rapid breathing (> 24 bpm), errors stayed low at 0.38 bpm and 0.49 bpm. When deliberate motion artifacts were introduced, standalone sensors showed large errors exceeding 3.8 bpm, whereas RespiraSense AI effectively muted corrupted frames, containing residual error to 0.95 bpm.

TABLE II
TUNED SYSTEM PARAMETERS AND OPERATIONAL THRESHOLDS

Module	Parameter	Tuned Setting	Engineering & Physiological Reason
Bandpass Filter	Cutoff Frequencies	0.10 Hz – 0.50 Hz	Covers 6 to 30 bpm; removes 1–2 Hz heartbeats and gravity baseline.
Audio Noise Gate	Threshold & Buffer	–42.0 dBFS, 2.0 s buffer	Prevents gate fluttering during natural pauses between breaths.
Quality Gate (SQI)	Acceptance Cutoff	SQI >= 0.70	Ensures signal energy is concentrated in true breathing frequencies.
Motion Classifier	Dynamic RMS Cutoff	0.40 m/s²	Distinguishes resting chest movement from talking or moving.
Welch PSD	Windowing & Overlap	10.0 s Hanning, 75% overlap	Balances clean frequency resolution with low response delay.
Personalized Norms	Baseline Window	7 Days (>= 70% coverage)	Accounts for patient-specific resting breathing baselines.

VI. PRACTICAL CONSIDERATIONS, GOVERNANCE & LIMITATIONS

A. Chest Movement Versus True Airflow: It is crucial to recognize that an accelerometer tracks the physical expansion of the chest, not the actual volume of air entering the lungs. In healthy breathing, chest expansion correlates well with tidal airflow. However, during obstructive sleep apnea (OSA), a patient's diaphragm contracts forcefully against a blocked throat, creating visible chest motion even though zero air reaches the lungs. Because of this, RespiraSense AI is intended as a respiratory rate tracker, not a tool for measuring lung volume or diagnosing obstructive hypopneas without separate airflow checks.

B. Data Privacy and Local Storage: Health information gathered in home environments requires careful privacy protection. RespiraSense AI keeps all patient demographics, telemetry logs, and 7-day baseline statistics stored on the local device in an encrypted SQLite database using Write-Ahead Logging (WAL). Patient and clinician accounts use role-based access tokens, and all hospital report exports are signed with SHA-256 hashes for data integrity.

C. Regulatory Status and Non-Diagnostic Prototype Use: RespiraSense AI is an engineering research prototype. It has not been reviewed by regulatory bodies such as the US FDA or European notified bodies as a certified medical device (Class II SaMD). The reports and rates generated by this prototype are meant to demonstrate tracking concepts and assist in clinical research, but they should never replace standard medical monitors or guide emergency medication adjustments.

VII. CONCLUSION & FUTURE WORK

In this project, we demonstrated that standard smartphones can provide reliable, non-invasive respiratory rate tracking when paired with careful signal conditioning, active noise gating, and adaptive sensor fusion. Across 1,200 observation epochs compared with gold-standard spirometry, RespiraSense AI achieved an overall error of 0.42 bpm, maintained tight agreement limits, and effectively suppressed body movement artifacts. By mathematically detailing the frequency limits of mobile accelerometers, we also clarified what smartphone sensors can and cannot detect.

Moving forward, we plan to validate the system in hospital sleep laboratories against full polysomnography (PSG) across larger patient groups, connect the logging system directly to hospital electronic health record (EHR) systems via FHIR APIs, and explore small on-device models to recognize abnormal breathing patterns like Cheyne-Stokes respiration.

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REFERENCES

[1] M. A. Cretikos, R. Bellomo, K. Hillman, J. Chen, S. Finfer, and A. Flabouris, "Respiratory rate: The neglected vital sign," *Med. J. Aust.*, vol. 188, no. 11, pp. 657–659, Jun. 2008.

[2] J. Subbe, R. Davies, E. Williams, P. Rutherford, and L. Gemmell, "Effect of introducing the Modified Early Warning score on clinical outcomes, early identification and resuscitation in patients at risk of physiological deterioration," *Resuscitation*, vol. 56, no. 3, pp. 267–274, Mar. 2003.

[3] M. A. F. Pimentel, A. E. W. Johnson, P. H. Charlton, M. Behar, et al., "Toward a robust estimation of respiratory rate from pulse oximeters," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 8, pp. 1914–1923, Aug. 2017.

[4] N. F. Liu, M. S. Tan, and J. R. Smith, "Continuous respiratory rate monitoring via wearable sensors in post-COVID-19 ambulatory cohorts," *Lancet Digit. Health*, vol. 3, no. 9, pp. e580–e588, Sep. 2021.

[5] F. J. Lovett, J. M. Buchan, and C. D. Smith, "Accuracy of observational respiratory rate recording on general medical wards," *BMJ Open*, vol. 9, no. 6, p. e029235, Jun. 2019.

[6] J. Folke, L. Cerina, M. Schmuecker, and T. Pehrson, "Critical evaluation of physiological methods for non-invasive respiratory monitoring in clinical environments," *Med. Biol. Eng. Comput.*, vol. 41, no. 4, pp. 377–383, Jul. 2003.

[7] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nat. Med.*, vol. 25, no. 1, pp. 44–56, Jan. 2019.

[8] M. Di Rienzo, E. Vaini, P. Castiglioni, et al., "Seismocardiography from smartphones: Feasibility, physiological significance, and comparative validation," *IEEE J. Biomed. Health Inform.*, vol. 22, no. 4, pp. 1042–1050, Jul. 2018.

[9] S. A. H. Al-shaer, A. A. Al-Haj, and M. I. Khalil, "Acoustic respiratory rate estimation using integrated smartphone microphones," *IEEE Sensors J.*, vol. 20, no. 16, pp. 9345–9353, Aug. 2020.

[10] P. H. Charlton, D. A. Birrenkott, T. Bonnici, et al., "Breathing rate estimation from the electrocardiogram and photoplethysmogram: A review," *IEEE Rev. Biomed. Eng.*, vol. 11, pp. 2–20, 2018.

[11] M. A. Pimentel, P. H. Charlton, and D. A. Clifton, "Probabilistic estimation of respiratory rate from wearable photoplethysmography," *Physiol. Meas.*, vol. 36, no. 7, pp. 1523–1547, Jul. 2015.

[12] J. Inverso, O. T. Inan, and G. T. Kovacs, "Continuous respiratory rate tracking via suprasternal seismocardiography," *IEEE Trans. Biomed. Circuits Syst.*, vol. 12, no. 3, pp. 589–598, Jun. 2018.

[13] A. S. Pandia, O. T. Inan, and J. A. G. M. Reiss, "Principal component analysis for multiaxial seismocardiographic respiration extraction," *IEEE J. Transl. Eng. Health Med.*, vol. 7, pp. 1–9, May 2019.

[14] B. A. Reyes, N. Reljin, and K. H. Chon, "Tracheal sounds acquisition using smartphone microphones for tidal volume and airflow monitoring," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 11, pp. 2732–2741, Nov. 2014.

[15] R. A. Al-Mously, M. M. Al-Akaidi, and S. M. Faris, "Acoustic identification of respiratory phases using spectral entropy and dynamic thresholds," *Comput. Biol. Med.*, vol. 112, p. 103362, Sep. 2019.

[16] A. Hannun, P. Rajpurkar, M. Haghpanahi, et al., "Cardiologist-level arrhythmia detection in ambulatory electrocardiograms using deep neural networks," *Nat. Med.*, vol. 25, no. 1, pp. 65–69, Jan. 2019.

[17] H. Pasterkamp, S. S. Kraman, and G. R. Wodicka, "Respiratory sounds: Advances in transmission, generation, and clinical computer analysis," *Am. J. Respir. Crit. Care Med.*, vol. 156, no. 3, pp. 974–987, Sep. 1997.

[18] A. R. Sovijarvi, F. Dalmasso, J. Vanderschoot, et al., "Definition of terms for computerised respiratory sound analysis and guidelines for lung sound recordings," *Eur. Respir. Rev.*, vol. 10, no. 77, pp. 585–605, 2000.